

Graph Evidence Is Not Text Evidence: Serialization Faithfulness as Structural Invariance in Graph-Grounded Language Models

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Abstract

Graph-grounded language models are usually evaluated by answer accuracy, yet a serialized prompt is a string representation of a graph rather than the graph itself. We argue that faithful graph-grounded reasoning requires the composed predictor $F_\theta \circ z$ to be well-defined on graph-query equivalence classes: structurally equivalent graph-query pairs should induce the same prediction regardless of textual realization. We formalize this as a structural invariance property, decompose unfaithfulness into syntactic instability, semantic interference, and distractor susceptibility, and introduce corresponding diagnostics. On controlled graph reasoning tasks, DeepSeek-V4-Flash achieves 98.9% accuracy and near-perfect syntactic stability ($FS = .975$), yet its isomorphism consistency drops from .969 to .701 under role-incongruent relabeling ($\Delta_{sem} = .268$); only 72.9% of originally correct answers survive this transformation. Across four model families, we observe targeted semantic interference in high-accuracy models, broad unfaithfulness in weaker models, and near-perfect invariance as a capability ceiling. A graph-grounded LLM can be accurate without being structurally faithful: its predictions may fail to remain invariant under graph-equivalent textual relabelings.

1 Introduction

Graph-grounded language models are used for graph reasoning, knowledge graph QA, and retrieval-augmented generation (Edge et al., 2025; He et al., 2024b; Tang et al., 2024). But these models consume text, not graphs; any graph evidence must pass through a *textual serialization* step, typically treated as a neutral implementation detail. We argue this is unwarranted. Citation faithfulness research establishes that an answer can be correct but not faithfully grounded in the provided evidence (Wallat et al., 2025); the same distinction applies to graph evidence. A model that answers

correctly by relying on node names rather than connectivity is correct but not robustly so.

This paper makes three contributions. First, we formalize serialization faithfulness as an invariance property over graph-query equivalence classes, establishing structural invariance as a necessary condition for graph faithfulness (Proposition 1). Second, we decompose unfaithfulness into syntactic instability, semantic interference, and distractor susceptibility. Third, we show that accuracy does not imply structural faithfulness: a model can answer correctly yet flip its answer under isomorphic relabeling with semantically misleading names.

2 Related Work

Graph-grounded LLMs. GraphGPT (Tang et al., 2024) aligns GNN-encoded tokens with LLM tokens; InstructGLM (Ye et al., 2024) encodes structure via language; GraCoRe (Yuan et al., 2025) notes “node ordering impacts task success”; TalkLikeaGraph (Fatemi et al., 2024) finds format-dependent accuracy variation; NLGraph (Wang et al., 2023) shows brittleness to spurious correlations. These establish that representation matters, but primarily evaluate accuracy. We isolate graph-preserving transformations as diagnostics for faithful interpretation.

Faithfulness and sensitivity. Citation correctness $\neq$ faithfulness (Wallat et al., 2025); input ordering can substantially affect LLM outputs (Guan et al., 2025); format sensitivity is a general LLM property (He et al., 2024a); in graph ML, permutation-invariant or permutation-equivariant computation is a fundamental requirement (Zhang et al., 2023). We connect these at the graph-to-language interface: does serialized prompting induce a well-defined graph reasoning function?

3 Serialization Faithfulness as Graph Invariance

A graph reasoning task defines a target function $y : \mathcal{G} \times \mathcal{Q} \rightarrow \mathcal{Y}$ over graph-query pairs. A serialization-based LLM instead computes a function over strings, $F_\theta : \Sigma^* \rightarrow \mathcal{Y}$. Serialization faithfulness asks whether $F_\theta \circ z$ is well-defined on graph-equivalence classes: equivalent graph-query pairs should induce the same prediction regardless of their textual realization.

Formal setup. A serialization map $z_{\rho,\pi,\ell} : \mathcal{G} \times \mathcal{Q} \rightarrow \Sigma^*$ depends on format ρ , ordering π , and labeling ℓ . *Structural equivalence*: $(G, q) \sim_s (G', q')$ if $G' = \phi(G)$ is an isomorphic relabeling and $q' = \phi(q)$. *Label-preserving equivalence*: $(G, q) \sim_\ell (G', q')$ if $y(G, q) = y(G', q')$. IC uses $\sim_s$; DR uses $\sim_\ell$. This distinction separates whether a model respects structural equivalence from whether it is robust to weaker answer-preserving perturbations. A model is serialization-faithful with respect to an equivalence $\sim$ if

$$(G, q) \sim (G', q') \Rightarrow F_\theta(z(G, q)) = F_\theta(z(G', q')) \quad (1)$$

Desideratum: well-defined graph reasoning under serialization. For a serialized LLM to behave as a graph-grounded reasoner, the composed predictor $F_\theta \circ z$ should be well-defined on graph-query equivalence classes. That is, if two serialized strings encode structurally equivalent graph-query pairs, then any difference in prediction reflects dependence on non-structural features of the serialization rather than on the graph evidence itself.

Proposition 1 (Structural invariance is necessary). Let h be a predictor whose value depends only on the isomorphism class of a graph-query pair (G, q) . Then for any isomorphism ϕ , $h(G, q) = h(\phi(G), \phi(q))$. Consequently, if $F_\theta \circ z$ changes its prediction under an isomorphic relabeling, the change cannot be explained by graph structure alone. *Proof.* An isomorphism preserves all structural relations (adjacency, paths, degrees, motifs). Since h depends only on the isomorphism class, relabeling identifiers cannot change its output; any prediction change must arise from non-structural features of the serialization.

Unfaithfulness decomposition. We decompose unfaithfulness into three components:

1. *Syntactic serialization instability* ($1 - \text{FS}$, OS): dependence on format or ordering.
2. *Semantic interference* ($\Delta_{\text{sem}} = \text{IC}_N - \text{IC}_M$): the gap between neutral and role-incongruent relabeling.
3. *Distractor susceptibility* ($1 - \text{DR}$): sensitivity to label-preserving irrelevant structure.

The central component is semantic interference, measured by isomorphism consistency under role-incongruent relabeling. Each diagnostic instantiates the same invariance requirement with a different transformation family: format stability varies the textual map z_ρ while holding the graph-query pair fixed; order sensitivity varies the ordering π ; isomorphism consistency varies the graph-query pair within structural equivalence $\sim_s$; distractor robustness varies the graph-query pair within label-preserving equivalence $\sim_\ell$. Thus all metrics test whether $F_\theta \circ z$ is well-defined under transformations that preserve the task-relevant graph evidence.

Isomorphism Consistency (IC). Whether structurally identical graphs with relabeled nodes yield the same answer: $\text{IC}(q, G, \phi) = 1[\hat{a}(q, z(G)) = \hat{a}(\phi(q), z(\phi(G)))]$. We test *neutral* relabeling ($A \rightarrow X_1, B \rightarrow X_2, \dots$) yielding IC_N , and *role-incongruent* relabeling yielding IC_M , where node names conflict with structural roles, testing whether the model follows connectivity when name semantics conflict with structure. We define:

$$\Delta_{\text{sem}} = \text{IC}_N - \text{IC}_M \quad (\text{semantic interference}) \quad (2)$$

$$\text{CF}_M = \Pr[\hat{a}_M = y \mid \hat{a}_0 = y] \quad (3)$$

CF_M measures whether predictions that were initially correct remain correct after semantic relabeling, a stricter test than consistency alone. $\text{CF}_M < 1$ indicates dependence on surface cues disrupted by the relabeling.

Supporting diagnostics. *Format Stability* (FS): agreement across formats; *Order Sensitivity* (OS): answer flip rate under reordering; *Distractor Robustness* (DR): whether label-preserving disconnected structure changes the prediction ($y(G, q) = y(G \cup D, q)$ required).

4 Experimental Setup

Tasks and graphs. Four graph reasoning tasks with structurally defined ground truth, on 100

Factor	Settings
Tasks	Connectivity, Shortest Path, Triangle, Common Nb.
Graphs	100 ER graphs, $n \in \{5..15\}$, $p=0.3$
Models	Qwen3.6-35B-A3B, DeepSeek-V4-Flash, Llama-4-Scout, gpt-5.4-mini, gpt-5.4 [†]
Formats	Edge list, Adj. list, JSON, NL
Relabeling	Neutral ($X_1, X_2, \dots$), Role-incongruent
Distractors	Disconn. comp., Δ , high-deg. star

Table 1: Experimental conditions. All distractors verified label-preserving. [†]gpt-5.4 results in Appendix A.

Model	Acc.	1-FS↓	OS↓	$\Delta_{\text{sem}} \downarrow$	1-DR↓	$\text{CF}_M \uparrow$
Symbolic oracle	1.00	.000	.000	.000	.000	1.00
gpt-5.4-mini	.936	.084	.046	.152	.066	.533
Qwen3.6-35B-A3B	.541	.325	.086	.061	.318	.575
DeepSeek-V4-Flash	.989	.025	.018	.268	.026	.729
Llama-4-Scout	.689	.259	.142	.091	.306	.658

Table 2: Main results. Columns match the unfaithfulness decomposition: syntactic instability (1-FS, OS), semantic interference (Δ_{sem}), distractor susceptibility (1-DR), and robust correctness (CF_M). The oracle computes each task from adjacency, ignoring node-name semantics. gpt-5.4 results in Appendix A.

Erdős-Rényi graphs ($n \in \{5..15\}$, $p = 0.3$), seeded deterministically, yielding 400 graph-task pairs.

Models. Three open-weight models across families (Qwen3.6-35B-A3B: Alibaba, 35B total/3B active, Apache-2.0; DeepSeek-V4-Flash: DeepSeek, 284B/13B active, MIT; Llama-4-Scout: Meta, 109B/17B active, Llama license) and a closed model (gpt-5.4-mini-2026-03-17); gpt-5.4-2026-03-05 as capability ceiling (Appendix A). All models use temperature 0; reasoning or thinking modes are disabled where configurable. Outputs not matching the task-specific answer schema count as incorrect for accuracy and as disagreements for consistency metrics.

Relabeling. Neutral relabeling uses generic identifiers ($X_1, X_2, \dots$); role-incongruent relabeling assigns semantic names contradicting structural roles (degree, leaf status, articulation, triangle participation). The synthetic graphs contain no semantic node attributes; these names appear only as textual identifiers.

5 Results

Table 2 presents the main results.

Accuracy does not imply faithfulness. gpt-5.4-mini achieves 93.6% accuracy, yet $\text{CF}_M = .533$: only 53.3% of correct answers survive

role-incongruent relabeling. DeepSeek-V4-Flash reaches 98.9% accuracy but still loses 27.1% of correct answers ($\text{CF}_M = .729$), with $\Delta_{\text{sem}} = .268$, the highest semantic interference observed. When surface semantic cues conflict with connectivity, predictions change despite the graph being preserved.

Example failure. In a shortest-path instance, the model correctly predicts distance 3 under neutral labels. Under isomorphic relabeling, a peripheral node is renamed Hub; the edges and correct answer are unchanged, yet the model predicts a shorter path, illustrating how surface role names can override graph connectivity (Δ_{sem}).

Three failure profiles. *Targeted semantic interference:* DeepSeek-V4-Flash has excellent syntax (1-FS = .025, OS = .018) and distractor robustness (1-DR = .026), but collapses under misleading names ($\text{IC}_N = .969 \rightarrow \text{IC}_M = .701$, $\Delta_{\text{sem}} = .268$). gpt-5.4-mini shows a similar but milder pattern ($\Delta_{\text{sem}} = .152$). *Broad unfaithfulness:* Qwen3.6-35B-A3B and Llama-4-Scout fail on all axes (1-FS = .325 / .259, 1-DR = .318 / .306), with small Δ_{sem} (.061 / .091) only because they are already unfaithful under neutral relabeling; their CF_M (.575 / .658) is partly sustained by the subset of easy tasks (connectivity) where relabeling matters less. For these lower-accuracy models, small Δ_{sem} should not be read as semantic robustness: when neutral consistency is already weak, the semantic gap is compressed by broad instability. The dominant failure mode shifts with capability: weaker models fail broadly, while stronger models fail specifically at semantic interference, and this failure is invisible to accuracy alone.

Capability ceiling. gpt-5.4 achieves near-perfect invariance ($\Delta_{\text{sem}} = 0$, $\text{CF}_M = 1.0$, FS = .989; Appendix A), suggesting faithfulness is capability-dependent. Notably, $\text{CF}_M < 1$ is possible even at $> 93\%$ accuracy, undetectable by accuracy alone.

6 Conclusion

We have shown that accuracy does not imply structural faithfulness. Proposition 1 establishes structural invariance as necessary; we decomposed unfaithfulness into syntactic instability, semantic interference, and distractor susceptibility. Across four model families, we observe three failure profiles: targeted semantic interference in high-accuracy models (DeepSeek: $\Delta_{\text{sem}} = .268$,

$CF_M = .729$ at 98.9% accuracy; gpt-5.4-mini: $\Delta_{sem} = .152$, $CF_M = .533$), broad unfaithfulness in weaker models (Qwen3.6, Llama-4-Scout), and near-perfect invariance as capability ceiling (gpt-5.4: $\Delta_{sem} = 0$). A graph-grounded LLM can be accurate without being structurally faithful.

Limitations

This study has several limitations. First, we evaluate on synthetic graphs of 5–15 nodes; effects may differ at scale. Second, our four formats do not cover all representations. Third, we extract answers via pattern matching. Fourth, we diagnose but do not propose improvement methods. Fifth, two models are proprietary; we include three open-weight models for broader model-family coverage.

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A Supplementary Results

gpt-5.4 (capability ceiling). gpt-5.4-2026-03-05: FS = 0.989, OS = 0.009, $IC_N = IC_M = 1.00$, $\Delta_{sem} = 0$, DR = 1.00, $CF_M = 1.0$.

Distractor Robustness & Order Sensitivity. DR: gpt-5.4-mini .934, gpt-5.4 1.000, DeepSeek .974, Qwen3.6 .682, Llama .694. OS: gpt-5.4-mini .046, gpt-5.4 .009, DeepSeek .018, Qwen3.6 .086, Llama .142.

Per-task Δ_{sem} . gpt-5.4-mini: .139/.277/.130/.064; DeepSeek: .247/.413/.157/.257; Llama: .020/.187/.000/.187; Qwen3.6: .060/.054/.077/.053 (connectivity/path/triangle/neighbor).

Model metadata. qwen/qwen3.6-35b-a3b (Apache-2.0), deepseek/deepseek-v4-flash (MIT), meta-llama/llama-4-scout (Llama license), gpt-5.4-mini-2026-03-17, gpt-5.4-2026-03-05 (proprietary). Accessed June 2026, temperature 0; reasoning disabled where configurable. Hosted inference may differ from local deployment.